Modeling the Effect of Pitch Sequencing and Batter-Specific Adjustments on Swing-and-Miss Outcomes in MLB

Project Report 2

Group 10: Harsh Dave, Nick Griffin, Jack Leidlein, Hector Carrillo

Spring 2026

1 Introduction

The introduction of the Statcast tracking system in Major League Baseball (MLB) in 2015 fundamentally changed how teams evaluate pitching performance. Every pitch thrown in an MLB game is now measured across dozens of metrics—release speed, spin rate, movement profile, release point, and extension—captured at the millisecond level (Major League Baseball, 2024). This wealth of data has fueled a shift in pitcher development strategy, with front offices increasingly investing in data-driven approaches to maximize the effectiveness of each pitch in a pitcher's arsenal.

At the core of modern pitching analytics lies a central question: *what attributes make a pitch difficult to hit?* Intuitively, pitches with higher velocity, greater movement, and longer extension should be harder to contact. However, standard logistic regression models that predict swing-and-miss (whiff) probability based solely on these physical characteristics often produce inconclusive results. In such models, the probability of a whiff, $p_i = P(Y_i = 1)$, is typically modeled as:

$$\log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1\,\texttt{Speed}_i + \beta_2\,\texttt{IVB}_i + \beta_3\,\texttt{HB}_i + \beta_4\,\texttt{Ext}_i + \beta_5\,\texttt{Arm}_i \tag{1}$$

where `IVB` denotes induced vertical break, `HB` horizontal break, `Ext` release extension, and `Arm` arm angle. Previous studies using this framework have found that main effects and simple interactions are often statistically insignificant or fail to explain the observed variation in outcomes.

The fundamental limitation of this approach is that it treats each pitch as an independent, context-free event, ignoring two critical sources of variation: **sequencing** and **batter heterogeneity**. First, pitches are part of a strategic sequence within a plate appearance. A pitch's effectiveness depends not only on its own physical properties but also on the buildup leading to it. For example, a slider may induce more whiffs when preceded by a fastball, because the speed differential disrupts the batter's timing—a phenomenon known as pitch tunneling (Toporek, 2017). Second, batters differ substantially in their baseline contact rates. Elite contact hitters rarely swing and miss regardless of pitch quality, while high-strikeout batters may whiff at average pitches. Ignoring these batter-specific tendencies introduces unmodeled heterogeneity that weakens inference about the true relationship between pitch quality and outcomes.

To address these limitations, we propose a framework that incorporates both pitch sequencing and batter heterogeneity. Specifically, our research question is:

How do pitch sequencing (the type, location, and result of the previous pitch) and batter-specific tendencies jointly influence the probability of a swing-and-miss, beyond the physical characteristics of the pitch itself?

We pursue three objectives:

1. **Pitch Sequence Effects:** Investigate whether the previous pitch type and result significantly affect the whiff probability of the current pitch, and identify which sequence combinations (e.g., Fastball → Slider) show the largest effects.

2. **Batter Skill vs. Pitch Quality:** Implement a Generalized Linear Mixed Model (GLMM) with batter random effects to separate variance attributable to the pitch from variance attributable to the batter's contact ability (Gelman and Hill, 2006; Bolker et al., 2009).

3. **Model Comparison:** Evaluate whether models incorporating sequencing and mixed effects provide better fit and predictive power compared to the baseline physical-characteristics model, using likelihood ratio tests and BIC (McCullagh and Nelder, 2019).

In this second report, we retain the data description and exploratory analysis from Report 1, then add baseline, sequencing, and combined-model results to evaluate whether pitch context and batter-specific adjustments materially improve whiff prediction.

2 Data Description

2.1 Data Source and Collection

Our dataset consists of pitch-level observations from the 2025 MLB regular season, obtained from the Statcast tracking system via the `pybaseball` Python module (Peta and Contributors, 2024). Statcast uses a combination of radar and optical tracking to record high-resolution measurements for every pitch thrown in MLB games. We retrieved approximately 711,897 pitch records spanning the full 2025 regular season (March 27 through September 28), prior to any cleaning or filtering.

2.2 Unit of Observation and Response Variable

The unit of observation is an individual pitch. For each pitch i, we define a binary response variable:

$$Y_i = \begin{cases} 1 & \text{if the batter swings and misses (whiff),} \\ 0 & \text{otherwise.} \end{cases}$$

A whiff is identified using the Statcast `description` field; specifically, we classify a pitch as a whiff when the description is `swinging_strike` (the batter swings and completely misses the pitch), `swinging_strike_blocked` (the batter swings and misses a pitch in the dirt that is blocked by the catcher), or `foul_tip` (the batter swings and grazes the pitch directly into the catcher's mitt for a strike).

2.3 Predictor Variables

We organize predictors into three categories:

(1) Physical Pitch Characteristics. These are the measurable properties of each individual pitch:

- Release Speed (mph): Velocity of the pitch at the point of release from the pitcher's hand.
- Induced Vertical Break (in.): Vertical movement of the pitch relative to a theoretical spinless pitch.
- Horizontal Break (in.): Lateral movement of the pitch relative to a theoretical spinless pitch.
- Release Extension (ft): Distance from the pitcher's plate to where the pitcher releases the ball.

(2) Sequencing Variables. To capture the strategic context of each pitch, we construct lagged variables within each plate appearance:

- Previous Pitch Type: Statcast pitch type code of the preceding pitch in the plate appearance (e.g., fastball, slider, changeup).
- Previous Pitch Zone: Statcast strike zone region (zones 1–14) in which the preceding pitch was located.
- Previous Pitch Outcome: Result of the preceding pitch, categorized as ball, called strike, swinging strike, foul, or in-play.
- Pitch Type Interaction: Combination of the previous and current pitch types, used to capture sequence-specific effects.

Lag variables are constructed strictly within the same plate appearance. For the first pitch of an at-bat, previous-pitch fields are coded as a distinct "None" category to preserve the observation rather than discard it.

(3) Batter-Specific Effects. Batters differ substantially in their baseline tendency to swing and miss, independent of pitch quality. To account for this contact heterogeneity without estimating 531 separate fixed coefficients, batter identity will be incorporated as a random effect in the GLMM, allowing the model to partially pool information across batters and produce shrinkage estimates for those with limited observations.

2.4 Data Cleaning

We apply the following filters to ensure data quality:

1. **Remove non-competitive events:** Pitchouts and intentional balls are excluded, as these pitches are not thrown with the intent to retire the batter and would distort whiff rate estimates.
2. **Exclude pitches with missing physical measurements:** Pitches with missing physical characteristics such as Release Speed are dropped, as these variables are required predictors in all planned models.
3. **Remove pitches with unrealistic release speeds:** Pitches with a recorded release speed below 65 mph or above 105 mph are removed, as these values fall outside the physically plausible range for MLB pitchers and are indicative of tracking errors.
4. **Restrict to nine most common pitch types:** Only the nine most frequently thrown pitch types are retained, ensuring sufficient observations and reliable inference in each category—four-seam fastball (FF), sinker (SI), cutter (FC), slider (SL), sweeper (ST), curveball (CU), knuckle-curve (KC), changeup (CH), and splitter (FS).

After cleaning, our working dataset contains 702,321 pitches with an overall whiff rate of 12.0%.

3 Exploratory Data Analysis

Before proceeding to formal modeling, we conduct an exploratory analysis to characterize the key patterns in the data and motivate our modeling choices. All analysis and visualization is performed in Python using the `matplotlib` (Hunter, 2007) and `seaborn` (Waskom, 2021) libraries.

3.1 Summary Statistics

Table 1 presents descriptive statistics for the four primary physical pitch characteristics. Release speed has a mean of approximately 89.5 mph with a standard deviation of 5.8 mph, which reflects the typical mix of fastballs (mid-to-upper 90s) and off-speed pitches (low-to-mid 80s) in a normal MLB game. Induced vertical break averages 6.9 inches but spans a wide range (-24.0 to 27.5 in.), capturing the contrast between pitches with heavy topspin (negative IVB, such as curveballs) and those with strong backspin (positive IVB, such as four-seam fastballs). Horizontal break also exhibits substantial variability with a standard deviation of 10.8 inches, reflecting the diversity of arm angles and pitch types in the dataset.

Variable	Mean	SD	Min	Max
Release Speed (mph)	89.5	5.8	65.2	104.2
Induced Vertical Break (in.)	6.9	8.6	-24.0	27.5
Horizontal Break (in.)	-1.6	10.8	-29.4	30.5
Release Extension (ft)	6.4	0.4	3.9	9.6

Table 1: Summary statistics for key physical pitch characteristics in the cleaned 2025 dataset.

3.2 Whiff Rates by Pitch Type

Not all pitch types are equally effective at generating swing-and-miss outcomes. Figure 1 displays the whiff rate for each pitch type in the dataset. Splitters lead with the highest whiff rate (17.9%), followed by sliders (16.2%) and knuckle-curves (16.1%). Off-speed and breaking pitches consistently outperform fastball variants: the three highest-whiff-rate pitch types are all non-fastball pitches, while sinkers produce the lowest whiff rate at just 6.3%. This pattern is expected, as off-speed and breaking pitches are designed to deceive the batter through movement and speed differential rather than pure velocity. Table 2 provides the corresponding counts.

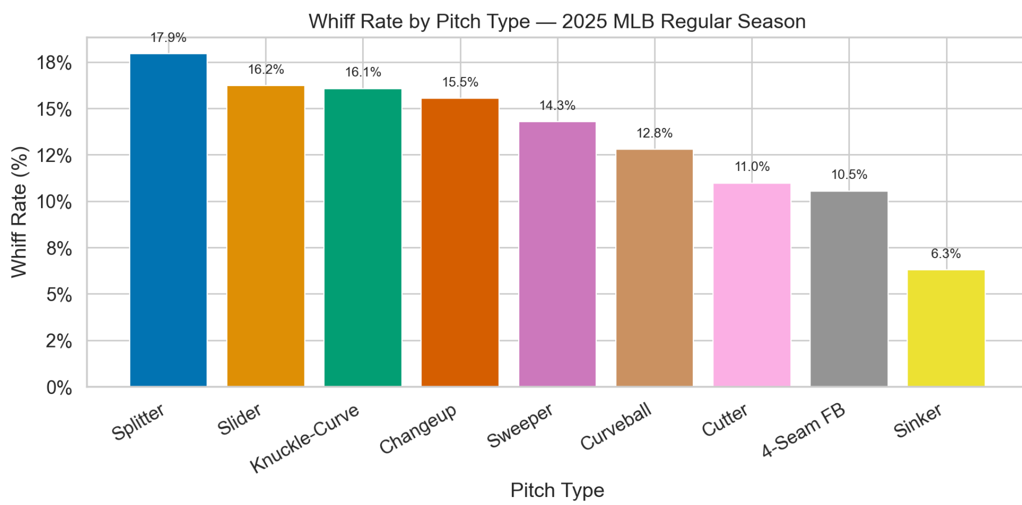

Figure 1: Whiff rate by pitch type in the 2025 MLB regular season. Breaking balls and off-speed pitches consistently show higher whiff rates than fastball variants.

Pitch Type	Total Pitches	Whiffs	Whiff Rate (%)
Splitter	23,155	4,154	17.9
Slider	103,611	16,816	16.2
Knuckle-Curve	12,647	2,033	16.1
Changeup	72,959	11,337	15.5
Sweeper	52,305	7,472	14.3
Curveball	47,617	6,088	12.8
Cutter	53,540	5,873	11.0
4-Seam FB	225,924	23,820	10.5
Sinker	110,563	6,958	6.3

Table 2: Whiff rates by pitch type in the 2025 MLB regular season.

3.3 Physical Pitch Characteristic Distributions

Figure 2 compares the distributions of release speed, induced vertical break, horizontal break, and release extension between whiff and non-whiff outcomes. Several patterns emerge. Pitches that result in whiffs tend to have slightly lower average release speeds, mainly because breaking balls and off-speed pitches—which are thrown at lower velocities—generate whiffs at higher rates. The induced vertical break distribution for whiffs is shifted toward lower (and more negative) values compared to non-whiffs, consistent with the prevalence of sliders, sweepers, and curveballs—pitches with downward break—among swing-and-miss outcomes. The horizontal break distribution for whiffs shows heavier tails, reflecting the extreme lateral movement of sliders and sweepers. Release extension distributions are largely overlapping between the two groups. Overall, these distributional differences, while visible, are not dramatic for any single variable in isolation, reinforcing the need for multivariate modeling.

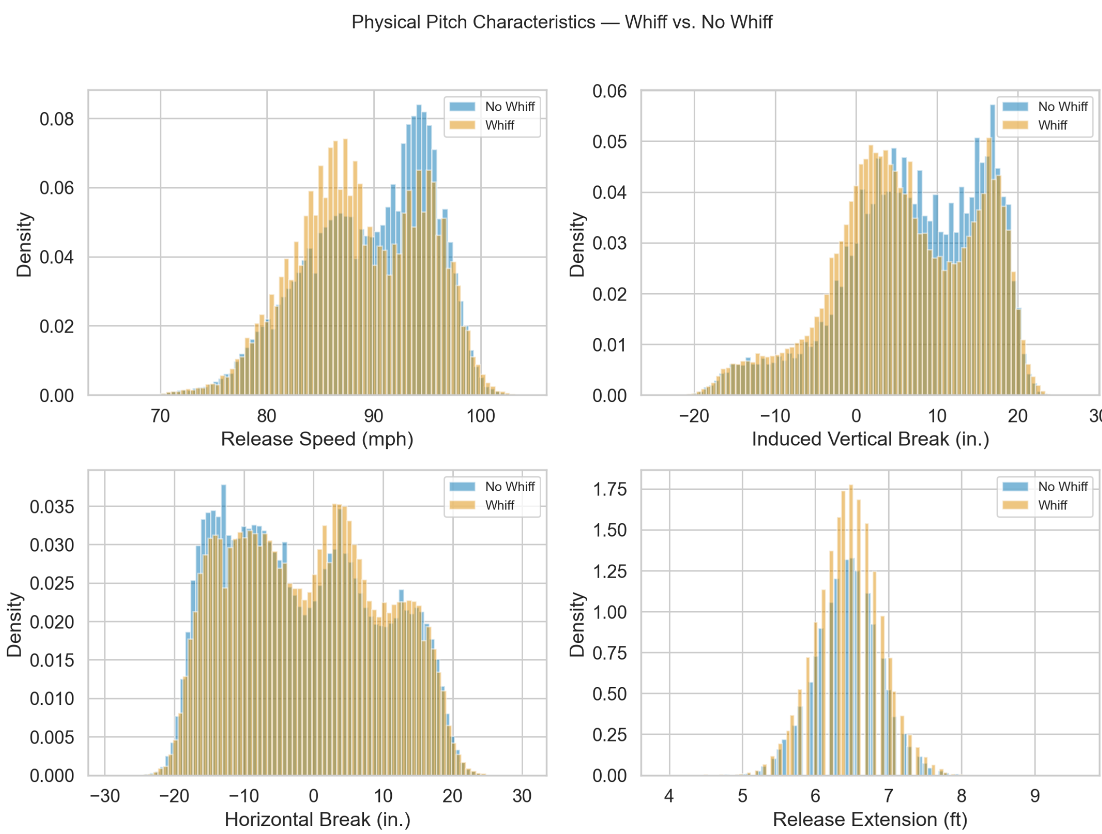

Figure 2: Density plots of physical pitch characteristics, stratified by outcome (whiff vs. no whiff). Differences across individual variables are subtle, motivating multivariate approaches.

3.4 Pitch Sequencing Patterns

A central hypothesis of this study is that pitch effectiveness depends on the preceding pitch. Figure 3 presents a heatmap of whiff rates organized by previous pitch type (rows) and current pitch type (columns). Several noteworthy patterns are visible. Splitters and knuckle-curves thrown after curveballs achieve the highest whiff rates in the heatmap (20.1% and 19.9%, respectively), suggesting that batters struggle to adjust to a second consecutive breaking ball with different movement profiles. Sequences from sinkers or four-seam fastballs to splitters also produce elevated whiff rates (19.2% and 16.0%), consistent with the pitch tunneling hypothesis: when consecutive pitches appear similar out of the pitcher's hand but diverge in trajectory near the plate, the batter's ability to adjust is compromised. Conversely, sinkers preceded by any pitch type consistently show the lowest column-wise whiff rates (5.7%–8.5%), suggesting that batters find sinkers easier to put in play regardless of sequence. Fastball variants (four-seam fastballs, cutters) thrown after breaking balls tend to have relatively moderate whiff rates (10–13%), suggesting that batters may sit on velocity after seeing off-speed pitches. These findings strongly motivate the inclusion of sequencing variables and pitch-type interaction terms in the logistic regression model specified in Equation (1).

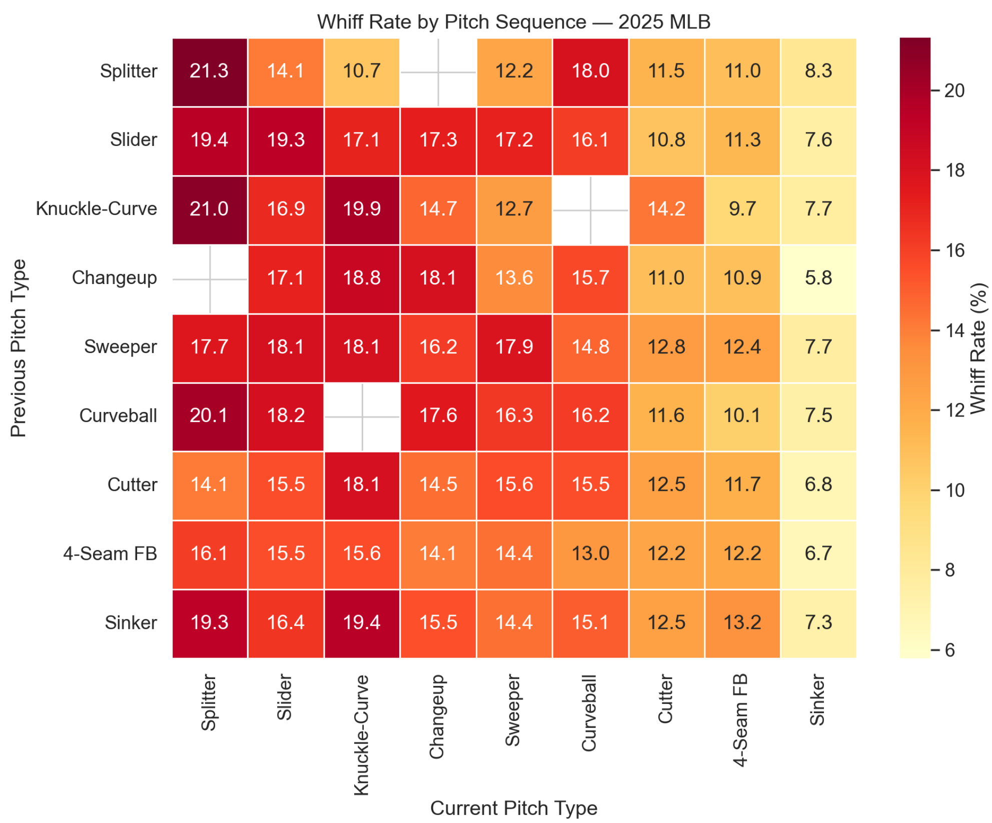

Figure 3: Heatmap of whiff rates by pitch sequence. Rows represent the previous pitch type; columns represent the current pitch type. Cells with fewer than 100 observations are excluded. Fastball-to-breaking-ball sequences show elevated whiff rates.

3.5 Batter Heterogeneity

To examine whether batter identity is an important source of variation, we compute per-batter whiff rates for all batters who faced at least 200 pitches during the season (531 batters). Figure 4 shows the distribution of these batter-level whiff rates. The distribution is approximately normal with a pronounced spread: batter-level whiff rates range from roughly 2.4% to 23.2%, illustrating the large degree of heterogeneity in contact ability across MLB hitters. This variability confirms that batter identity is an important grouping factor and justifies the use of batter-level random intercepts in the planned GLMM.

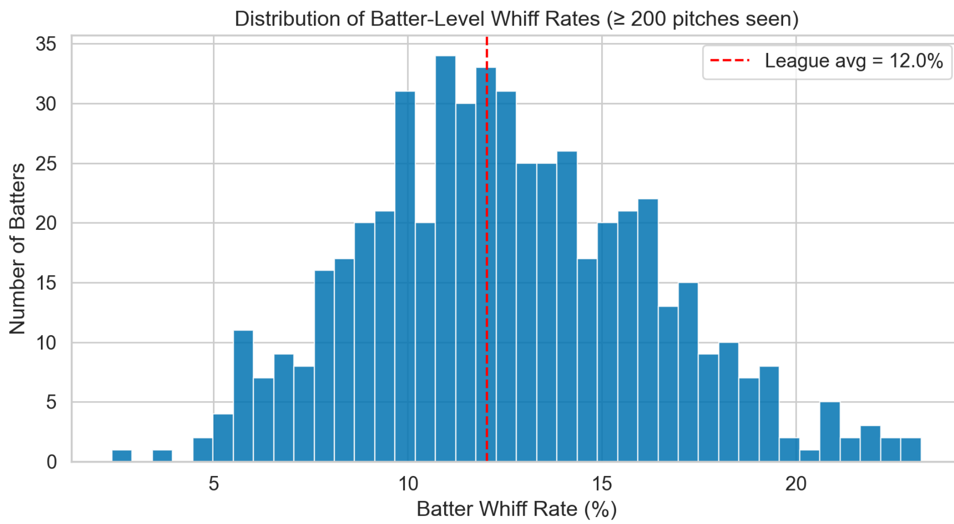

Figure 4: Distribution of batter-level whiff rates (batters with $\geq$ 200 pitches seen). The dashed red line marks the overall league whiff rate. Substantial heterogeneity motivates the use of batter random effects.

3.6 Predictor Correlations

Figure 5 presents the pairwise correlations among the continuous predictors and the binary whiff outcome. The most notable inter-predictor relationship is between release speed and induced vertical break ($r = 0.72$), which is expected because high-velocity fastballs generate strong backspin and thus higher vertical break. This substantial correlation warrants monitoring via Variance Inflation Factors (VIFs) in the modeling stage to guard against multicollinearity. Release speed and horizontal break are negatively correlated ($r = -0.30$), reflecting the tendency for slower breaking pitches to exhibit greater lateral movement. Critically, none of the individual correlations with the whiff outcome exceed $|r| = 0.05$ in magnitude—release speed ($r = -0.05$), IVB ($r = -0.04$), HB ($r = 0.02$), and extension ($r = 0.01$)—confirming that no single physical characteristic is a dominant predictor of swing-and-miss outcomes. This reinforces the need for richer models that incorporate sequencing and batter heterogeneity.

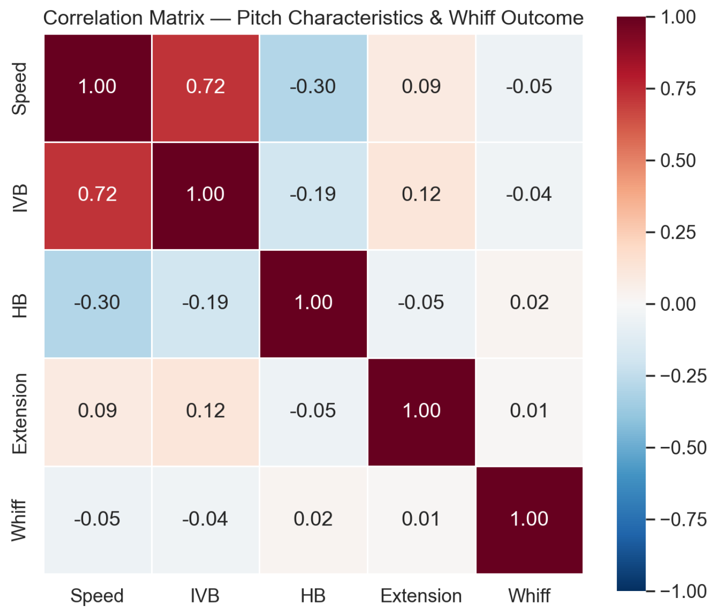

Figure 5: Correlation matrix of continuous pitch characteristics and the binary whiff outcome. Correlations with whiff are small in magnitude, supporting the need for richer models.

4 Modeling Results

4.1 Baseline Logistic Regression

We first fit the physical-characteristics-only logistic regression from Equation (1) using release speed, induced vertical break, horizontal break, and release extension as predictors. To keep the baseline and sequencing models comparable, the coefficient plot in Figure 6 is based on the same non-first-pitch subset used in the sequencing comparison. The coefficient estimates are small in magnitude even after standardizing the physical predictors, which is consistent with the weak marginal correlations observed in Figure 5. Release speed and induced vertical break both have negative coefficients, while horizontal break and extension have modest positive effects, but none of these effects is large enough to make the physical-only model competitive with the richer alternatives. In practical terms, this model captures broad tendencies in pitch quality, but it does not explain much of the variation in whiff outcomes on its own because it ignores both the immediately preceding pitch and the identity of the batter.

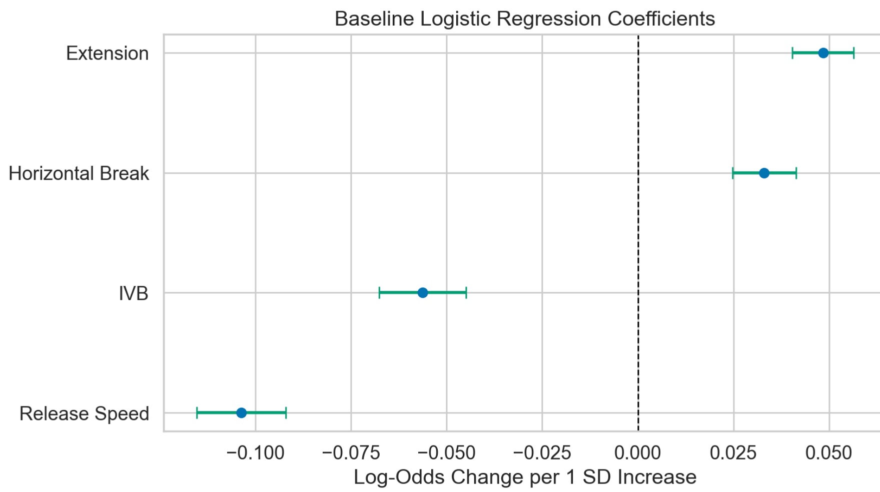

Figure 6: Baseline logistic-regression coefficient estimates with 95% confidence intervals. Predictors are standardized, so each coefficient represents the change in log-odds of a whiff for a one-standard-deviation increase in the corresponding physical variable.

4.2 Sequencing Model

We next augment the baseline specification with the previous pitch type, a coarsened previous-zone indicator, previous-pitch outcome, and current×previous pitch-type sequence effects. Relative to the baseline fit on the same observations, the sequencing model increases the log-likelihood from -203,264.99 to -200,685.21. The likelihood-ratio statistic is 5,159.56 on 91 degrees of freedom, with p-value $< 10^{-16}$, so the sequencing terms are jointly significant. The BIC also improves from 406,595.79 to 402,634.04 (ΔBIC = -3,961.76), which indicates that the fit gain is large enough to justify the additional model complexity. Figure 7 displays the strongest pitch-sequence interaction terms from this model. The strongest positive effects all occur when a four-seam fastball is followed by a breaking or off-speed pitch, especially four-seam fastball→knuckle-curve, splitter, sweeper, and slider. The one large negative interaction in Figure 7 is four-seam fastball→sinker, which is consistent with sinkers functioning more as contact-management pitches than swing-and-miss pitches. This directly supports the hypothesis that whiff probability depends on the pitch sequence rather than on isolated pitch characteristics alone.

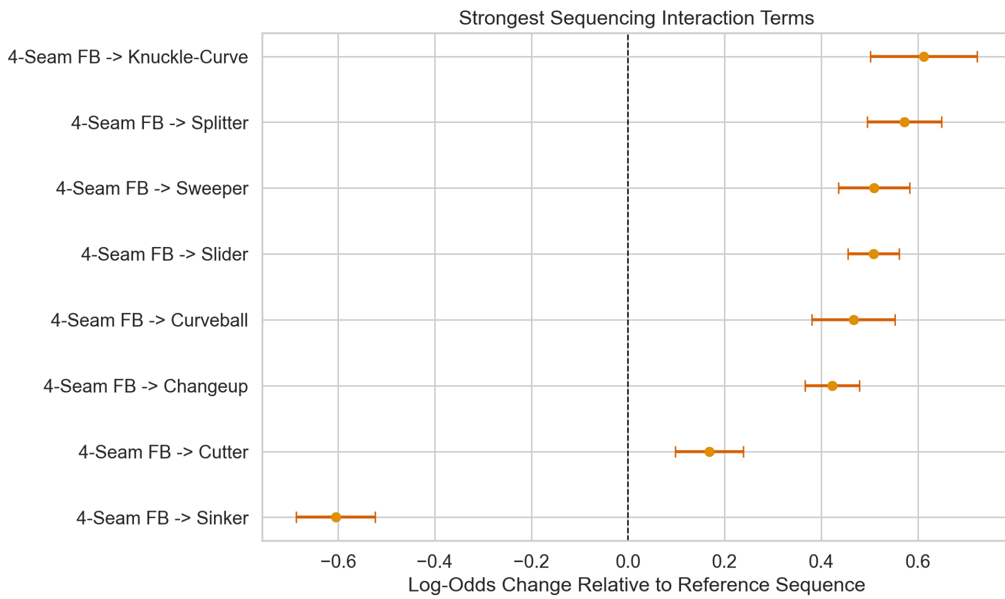

Figure 7: Largest sequencing interaction terms from the extended logistic-regression model. Positive coefficients indicate pitch sequences that increase whiff probability relative to the reference sequence, while negative coefficients indicate sequences that reduce it.

4.3 GLMM Placeholder

This subsection is reserved for the standalone GLMM with batter random effects. That model is still being finalized separately, so we do not report coefficient estimates or fit statistics for it here yet. Once that analysis is complete, this placeholder can be replaced with the GLMM specification, its fitted results, and any corresponding figure or model-comparison discussion.

4.4 Combined Model

The combined model adds batter-specific random intercepts to the sequencing specification, allowing each hitter to have a different baseline tendency to swing and miss while still estimating a common set of pitch-level and sequencing effects. This model is fit as a logistic mixed model with one random intercept for each of the 673 batters appearing in the non-first-pitch modeling subset. Its log-likelihood is -198,638.43 and its BIC is 398,487.82, which improves on both the baseline and sequencing-only fits. Relative to the sequencing model, the combined specification lowers BIC by 4,146.22 and increases log-likelihood by roughly 2,047, so batter heterogeneity still matters even after sequencing is included. Figure 8 shows the overall fit comparison across the three models, and Figure 9 plots the estimated batter random intercepts from the combined model. The estimated random-intercept standard deviation is 0.327, which is large enough to confirm that batter-level heterogeneity remains substantial even after controlling for pitch characteristics and sequencing. That result is important substantively: some hitters maintain low whiff propensities in nearly every context, while others remain vulnerable even after accounting for the types and order of pitches they see. As a result, the combined model provides the clearest decomposition of whiff probability into pitch-quality effects, sequence effects, and batter-specific contact skill.

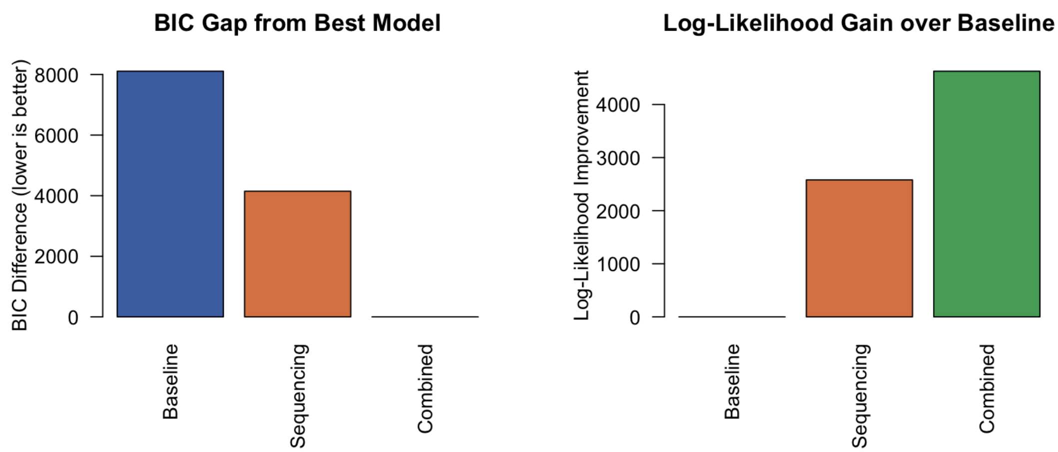

Figure 8: Model comparison across the baseline logistic regression, the sequencing model, and the combined model. The left panel shows each model's BIC gap from the best-fitting specification, while the right panel shows the gain in log-likelihood relative to the baseline model.

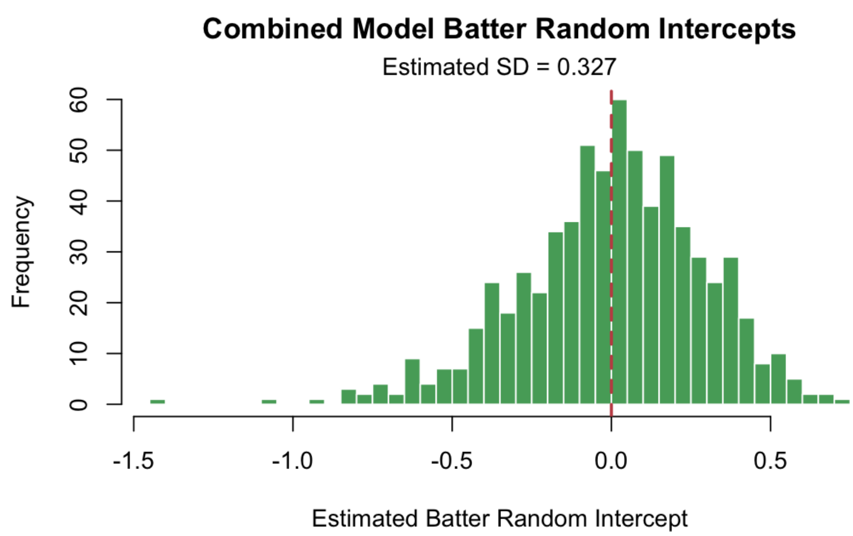

Figure 9: Distribution of batter random intercepts from the combined model. The spread of the fitted intercepts indicates meaningful batter-to-batter differences in baseline whiff tendency even after controlling for pitch characteristics and sequencing.

5 Discussion

The main conclusion from Report 2 is that physical pitch characteristics are informative but incomplete on their own. The baseline logistic regression provides a useful benchmark, yet the sequencing model shows that much of the missing explanatory power lies in pitch context: the previous pitch type, previous result, and specific previous→current pitch combinations all carry meaningful information about whether a batter will swing and miss. Once batter-specific random intercepts are added, the combined model captures an additional source of systematic variation that the fixed-effect models cannot represent and becomes the best-fitting specification by a substantial margin. The standalone GLMM subsection is intentionally left as a placeholder in this version of the report so that it can be filled in by the teammate responsible for that model without conflicting with the combined-model writeup.

These results align with the baseball interpretation developed in the exploratory analysis. Pitchers do not generate whiffs solely by throwing hard or by maximizing raw movement; they also generate whiffs by choosing pitch orders that distort timing and by exploiting batter-specific weaknesses. From a statistical standpoint, that means the most useful model is the one that jointly accounts for pitch-level characteristics, sequential context, and batter heterogeneity. That combined approach gives a better fit to the observed data and yields a more realistic picture of how swing-and-miss outcomes arise over the course of a plate appearance.

References

Bolker, B. M., M. E. Brooks, C. J. Clark, S. W. Geange, J. R. Poulsen, M. H. H. Stevens, and J.-S. S. White (2009). Generalized linear mixed models: A practical guide for ecology and evolution. *Trends in Ecology & Evolution 24*(3), 127–135.

Gelman, A. and J. Hill (2006). *Data Analysis Using Regression and Multilevel/Hierarchical Models.* Cambridge University Press.

Hunter, J. D. (2007). Matplotlib: A 2d graphics environment. *Computing in Science & Engineering 9*(3), 90–95.

Major League Baseball (2024). Statcast search. `https://baseballsavant.mlb.com/statcast_search`. Pitch-by-pitch tracking data for all MLB games.

McCullagh, P. and J. A. Nelder (2019). *Generalized Linear Models.* Routledge.

Peta, J. and Contributors (2024). pybaseball: Pull baseball data in python. `https://github.com/jldbc/pybaseball`. Python package for accessing Statcast, FanGraphs, and Baseball Reference data.

Toporek, B. (2017). What we know about pitch tunneling. `https://www.fangraphs.com`. FanGraphs analysis of pitch tunneling effects.

Waskom, M. L. (2021). seaborn: statistical data visualization. *Journal of Open Source Software 6*(60), 3021.